WORKED SOLUTION

STEP III 2006, Question 13

Probability and Statistics • Geometric Probability

STEP III 2006, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/06-s3-q13

The key idea

The perimeter is made of two very different pieces, an arc of length π and a diameter of length 2, so we condition on which piece each point lands on. In each of the three cases the area of the triangle has a simple formula, $\frac{1}{2}ab\sin C$ with the vertex C at the centre, and the conditional expectations are short integrals. The law of total expectation then puts the pieces back together.

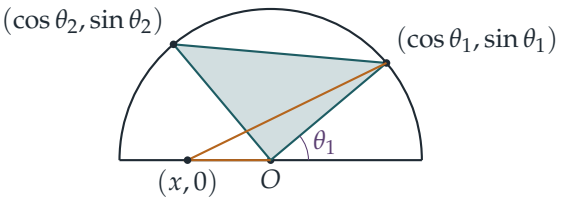
Solution

Setting up

Put the centre O of the diameter at the origin, with the diameter along the x-axis from $(-1,0)$ to $(1,0)$ and the arc in the upper half-plane. "At random on the perimeter" means uniformly distributed by length along the whole boundary, whose total length is $\pi+2$. So each point independently

- lies on the arc with probability $\dfrac{\pi}{\pi+2}$, and then it is $(\cos\theta,\sin\theta)$ with θ uniform on $[0,\pi]$;
- lies on the diameter with probability $\dfrac{2}{\pi+2}$, and then it is $(x,0)$ with x uniform on $[-1,1]$.

(Uniform by length on the arc is the same as uniform in angle, because arc length on a unit circle equals the angle.)



Let D be the number of the two points that lie on the diameter. Then $D = 2, 1, 0$ with probabilities

$$\frac{4}{(\pi+2)^2}, \qquad \frac{2\cdot 2\pi}{(\pi+2)^2} = \frac{4\pi}{(\pi+2)^2}, \qquad \frac{\pi^2}{(\pi+2)^2},$$

using independence of the two points (the factor 2 in the middle case counts which point is on the diameter). In every case the triangle has a vertex at O, so its area is $\frac{1}{2}\,|OP|\,|OQ|\sin\angle POQ$.

Both points on the diameter

All three vertices are on a line, so the triangle is degenerate and $A = 0$. Hence $\mathrm{E}(A \mid D = 2) = 0$.

One point on the arc, one on the diameter

Say the points are $P = (\cos\theta, \sin\theta)$ and $R = (x, 0)$. Then $|OP| = 1$, $|OR| = |x|$, and the base OR lies along the diameter while P is at height $\sin\theta \geq 0$ above it, so

$$A = \tfrac{1}{2}\,|x|\sin\theta.$$

Since x and θ are independent,

$$\mathrm{E}(A \mid D = 1) = \tfrac{1}{2}\,\mathrm{E}|x|\,\mathrm{E}(\sin\theta) = \tfrac{1}{2}\cdot\frac{1}{2}\int_{-1}^{1}|x|\,\mathrm{d}x\cdot\frac{1}{\pi}\int_0^{\pi}\sin\theta\,\mathrm{d}\theta = \tfrac{1}{2}\cdot\tfrac{1}{2}\cdot\frac{2}{\pi} = \frac{1}{2\pi}.$$

Both points on the arc

Now $P = (\cos\theta_1, \sin\theta_1)$ and $Q = (\cos\theta_2, \sin\theta_2)$ with θ_1, θ_2 independent and uniform on $[0, \pi]$. Both radii have length 1 and the angle POQ is $|\theta_1 - \theta_2|$, so

$$A = \tfrac{1}{2}\,|\sin(\theta_1 - \theta_2)|.$$

? **Ask yourself** Do we need the distribution of $\theta_1 - \theta_2$ to find $\mathrm{E}|\sin(\theta_1 - \theta_2)|$?

No: condition on θ_2 first. For a fixed θ_2, as θ_1 runs over $[0, \pi]$ the angle $\theta_1 - \theta_2$ runs over an interval of length π. The function $|\sin u|$ has period π, so its integral over *any* interval of length π is the same as over $[0, \pi]$:

$$\int_0^{\pi}|\sin(\theta_1 - \theta_2)|\,\mathrm{d}\theta_1 = \int_0^{\pi}\sin u\,\mathrm{d}u = 2.$$

So $\mathrm{E}\big(|\sin(\theta_1-\theta_2)| \,\big|\, \theta_2\big) = 2/\pi$ whatever θ_2 is, and therefore

$$\mathrm{E}(A \mid D=0) = \tfrac{1}{2}\cdot\frac{2}{\pi} = \frac{1}{\pi}.$$

Putting it together

By the law of total expectation,

$$\begin{aligned}\mathrm{E}(A) &= \frac{4}{(\pi+2)^2}\cdot 0 + \frac{4\pi}{(\pi+2)^2}\cdot\frac{1}{2\pi} + \frac{\pi^2}{(\pi+2)^2}\cdot\frac{1}{\pi}\\ &= \frac{2+\pi}{(\pi+2)^2} = \frac{1}{2+\pi},\end{aligned}$$

which is $(2+\pi)^{-1}$, as required.

Checks

- The three case probabilities $\dfrac{4+4\pi+\pi^2}{(\pi+2)^2}$ add to 1.
- Each conditional expectation is at most the largest possible area in that case: $\frac{1}{\pi} \approx 0.318 \le \frac{1}{2}$ (two radii at right angles) and $\frac{1}{2\pi} \approx 0.159 \le \frac{1}{2}$.
- A Monte Carlo simulation with 10^6 pairs of points gave $\mathrm{E}(A) \approx 0.1943$, against $1/(2+\pi) \approx 0.1945$; quadrature confirmed the conditional values $1/\pi$ and $1/(2\pi)$.

Common slips

- Choosing each point on the arc or the diameter with probability $\frac{1}{2}$ each. "At random on the perimeter" weights each piece by its length.
- Forgetting the factor 2 in $\mathrm{P}(D=1)$: either point may be the one on the diameter.
- Writing $A = \frac{1}{2}\sin(\theta_1-\theta_2)$ without the modulus, which gives expectation 0 by symmetry.

Beyond STEP

This is a small piece of *geometric probability*, whose classic examples include Buffon's needle and Bertrand's paradox. Bertrand's paradox shows why the phrase "at random" must be pinned down, exactly as we did by taking length along the perimeter as the measure. The tool doing the work here, conditioning on a partition and averaging (the law of total expectation), is central to first-year probability. Grimmett and Welsh, *Probability: An Introduction*, covers it well.

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